# Vaibhav Patil

*AI/ML Engineer*

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## PROFILE

- AI Engineer specializing in building and deploying production-grade AI systems, including LLM-powered analytics platforms, genAI powered chatbots, and RAG solutions, with a strong focus on scalable deployment and business impact.

## PROFESSIONAL EXPERIENCE

**AI Engineer** 05/2026 – Present | remote

*Quantal AI*

- **Architected and developed the backend** for an AI-powered voice automation platform that converts natural-language deal discussions from meetings and calls into structured business entities such as **tasks, opportunities, and contacts** for downstream workflow automation, achieving 100% entity recall in evaluation, with ~70% of extractions requiring no field-level correction
- **Designed and implemented the FastAPI backend and relational data model** using PostgreSQL/Cloud SQL, SQLAlchemy, and Alembic, covering conversations, messages, voice memos, background jobs, and API request tracking.
- Built an **end-to-end voice processing pipeline** using **OpenAI APIs and Whisper**, with asynchronous background workers for processing **10–20 minute recordings** and LLM-based extraction of structured business information.
- **Deployed and maintained the backend on GCP Cloud Run**, using **Google Cloud Storage** for audio recordings, Cloud SQL for persistent data, and **GitHub Actions for CI/CD**; designed the architecture for human-in-the-loop intent confirmation and downstream Salesforce/Trello automation.
- **Architected a multi-agent voice AI system** using **Retell AI, n8n, and GoHighLevel** to automate patient calls including appointment booking, rescheduling, cancellations, and insurance queries.
- Implemented **intent-based workflow routing and CRM automation**, synchronizing patient information and triggering **SMS/email confirmations** through integrated business workflows.

**AI/ML Engineer** 11/2024 – 04/2026 | Mumbai, India

*AVAXPRO TECH PVT LTD*

- Designed and implemented a **Gen AI reporting platform using RAG** pipeline enabling natural-language analytics over complex **PostgreSQL** schema using a custom semantic layer, intelligent table detection, and role based access control **LLM-driven text to SQL generation (GPT-4o-mini on Amazon Bedrock)**.
- Built a **semantic embedding layer using PGVector** defining business-friendly table and column mappings, aliases, data types, and aggregation rules, enabling accurate **NL-to-SQL translation**, correct aggregations, and elimination of common semantic query errors.
- Developed and deployed an end-to-end **document intelligence pipeline** on **AWS (EC2, S3)** using **PyMuPDF, OCR, and LLM-based schema-driven extraction**, converting large unstructured PDFs into structured **Excel/CSV** datasets for downstream e-commerce systems.

**AI/ML intern** 04/2024 – 10/2024 | Mumbai, India

*AVAXPRO TECH PVT LTD*

- Engineered an **LLM-powered RAG chatbot** on internal company data using semantic **embeddings**, vector databases (**Pinecone, ChromaDB**), and optimized similarity search, achieving improved intent accuracy and a 55% CSAT score.
- Designed, containerized, and deployed a **FastAPI-based chatbot** using Docker on AWS EC2, improving scalability and reducing deployment-related issues by 40%.
- Built and executed **ETL and data analysis pipelines** using pandas to clean, transform, and analyze large-scale sales, purchase, and stock datasets for business insights.

## SKILLS

- Programming & Backend : Python, FastAPI, Docker, GitHub, AWS(S3, EC2,Amazon Bedrock), GCP
- Databases & Data : MySQL, PostgreSQL, pandas, Apache Airflow, Pinecone, ChromaDB, PGVector
- AI & Machine Learning: Agentic AI, genAI, MCP , NLP, LLMs, RAG, Hugging Face, Scikit-learn, Vector Search, Ollama, Langchain, Retell AI, Voice AI , Langraph, ElevenLabs, Vapi, A2A, Whisper

## EDUCATION

**Masters in Information Technology** 07/2023 – 07/2025 | Mumbai
*Mumbai University*
CGPA: 8.61

**Bachelor in Information Technology** 06/2020 – 06/2023 | Mumbai
*Mumbai University*
CGPA : 8.95